



Review

Dynamic modeling of gearbox faults: A review

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ABSTRACT

Gearbox is widely used in industrial and military applications. Due to high service load, harsh operating conditions or inevitable fatigue, faults may develop in gears. If the gear faults cannot be detected early, the health will continue to degrade, perhaps causing heavy economic loss or even catastrophe. Early fault detection and diagnosis allows properly scheduled shutdowns to prevent catastrophic failure and consequently result in a safer operation and higher cost reduction. Recently, many studies have been done to develop gearbox dynamic models with faults aiming to understand gear fault generation mechanism and then develop effective fault detection and diagnosis methods. This paper focuses on dynamics based gearbox fault modeling, detection and diagnosis. State-of-art and challenges are reviewed and discussed. This detailed literature review limits research results to the following fundamental yet key aspects: gear mesh stiffness evaluation, gearbox damage modeling and fault diagnosis techniques, gearbox transmission path modeling and method validation. In the end, a summary and some research prospects are presented.

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1. Introduction

In a gearbox, gears and gear trains are used to provide speed and torque conversions from a rotating power source to another device. It is widely used in industrial, civilian and military applications, for example in helicopters, wind turbines, bucket wheel excavators, tracked loaders, and milling machines as shown in Fig. 1. In industrial applications, gearboxes may work under constant operation condition or varying operation condition. This paper focuses on reported studies on the constant operating condition. Limited work in the area of dynamic modeling has been reported on the varying operating condition.

According to the arrangement of gear wheels, gear trains can be classified into four categories [2]: simple gear train, compound gear train, reverted gear train and planetary gear train (epicyclic gear train). One example is given in Fig. 2 for each type of gear train. The simple gear train has one gear mounted on each shaft. If there is more than one gear mounted on a shaft, the gear train is called the compound gear train. If the axes of the driving gear shaft and the driven gear shaft are co-axial, the gear train is known as reverted gear train. If one gear rotates on its own axis and also revolves around the axis of another gear, this gear train is termed as planetary gear train. A total of 34 different types of planetary gear sets are described in Ref. [3]. A basic planetary gear set contains one sun gear, one internal gear (ring gear), one carrier and several planet gears that mesh with the sun gear and the ring gear simultaneously. The first three gear trains are collectively called fixed-axis gear trains in this paper since all gears only rotate on their own axis and all their axes are fixed. Comparing with fixed-axis gearboxes, planetary gearboxes can afford higher torque load due to the load sharing among multiple gear pairs and



Fig. 1. Some applications of gearboxes [1].

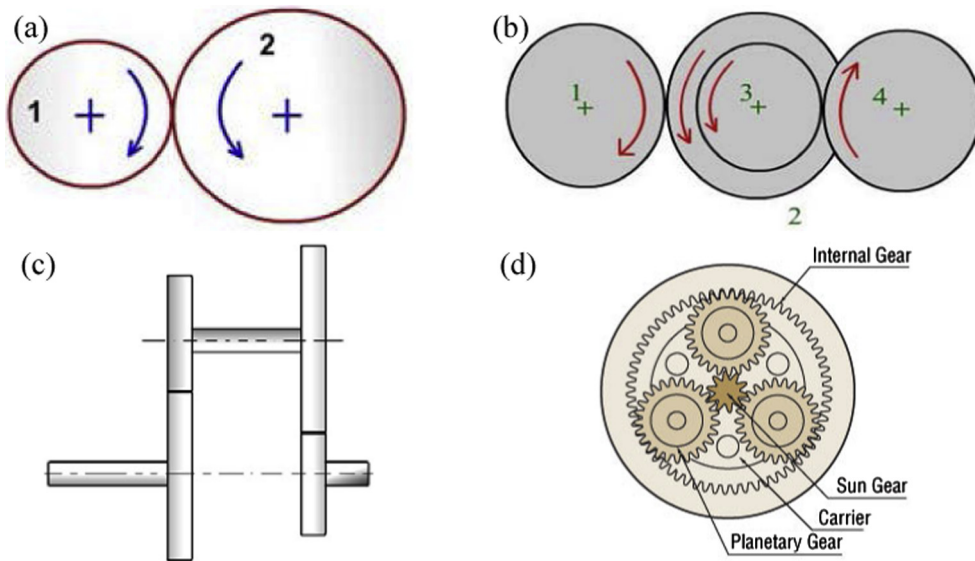


Fig. 2. Four types of gear trains: (a) simple gear train, (b) compound gear train, (c) reverted gear train and (d) planetary gear train.

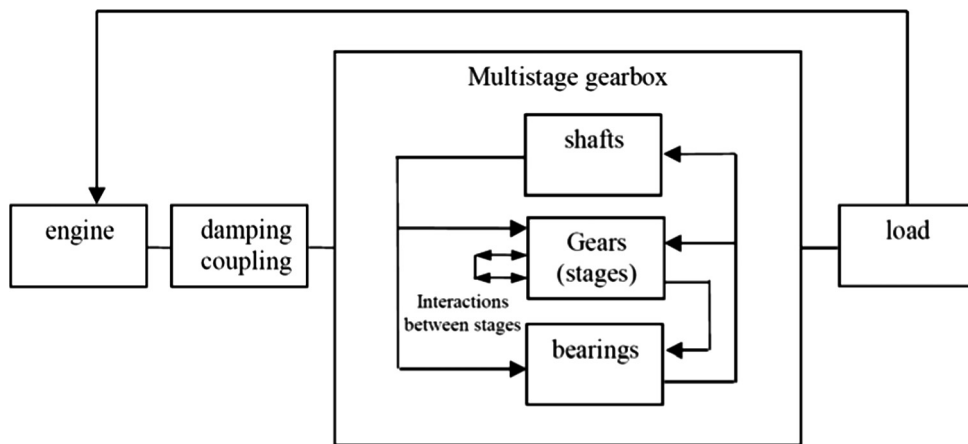


Fig. 3. A multistage gearbox and component interaction [4].

generate larger transmission ratio with equal or smaller volume [1]. However, the design and analysis of planetary gearboxes are generally harder than that of fixed-axis gearboxes as multiple sun-planet gear pairs and ring-planet gear pairs are meshing simultaneously in a planetary gearbox.

In heavy machinery, it is common to have complex multistage gearboxes [4]. Each stage consists of gears, bearings, shafts and supported structure as shown in Fig. 3. Bartelmus [4] proposed the concept of driving schemes including system of elements (engine, gear, bearing, shaft etc.), influencing factors (design, production, operation and condition change), interaction of gearbox elements and also with the working environment, and component damage mechanism. Understanding the factors in the driving schemes may help researchers in developing condition monitoring and diagnostic methods [4].

Due to high service load, harsh operating conditions or simple fatigue, faults may develop in gears [5,6]. If the gear faults cannot be detected early, the health will continue to degrade, perhaps causing large economic loss or catastrophe. In 2009, a helicopter crashed into the North Sea. All 14 offshore workers and two crewmen died. This accident was caused by the failure of the main rotor gearbox due to a gear fatigue crack [7]. Therefore, it is important to monitor the health of gearbox systems and detect faults as early as possible. Early detection allows proper scheduled shutdown and maintenance to prevent catastrophic failure, and consequently guarantees a safer operation and higher cost reduction [8].

Based on our literature review, there are mainly two types of approaches for gear fault diagnostics: data-driven based methods and physical model-based methods. Data-driven based methods purely rely on the analysis of historical data collected from gearboxes to diagnose and/or predict their health conditions. The measured data could be vibration signals,

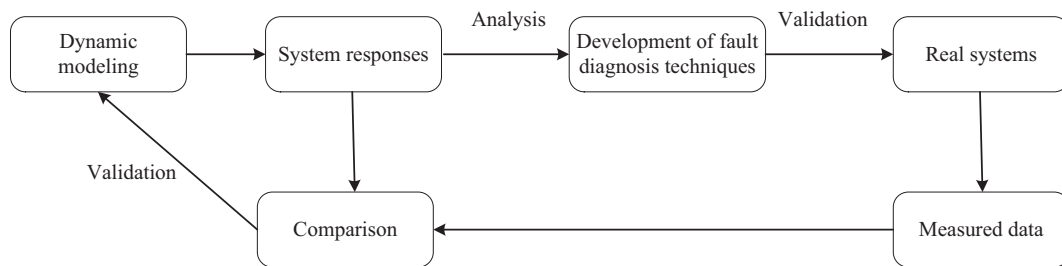


Fig. 4. Dynamics based fault diagnostics.

motor voltage and current signals, torque load signals, acoustic signals, metal scan data, gear weight loss data, gearbox strain signals, gear damage images and so on. Many data-driven methods were developed for gearbox fault diagnostics [9–19]. Physical model based methods built a virtual system to mimic an existing object based on human understanding of this object. The physical models of gearboxes can be divided into two subsets: modulation based models and dynamics based models. Modulation based models are developed via the understanding of amplitude modulation, frequency modulation and phase modulation characteristics of vibration signals. The studies on the development of modulation based models are available in Refs. [20–25]. Dynamics based models are developed based on a fundamental analysis of gear mesh mechanism and dynamics, then, dynamic characteristics in various health conditions can be simulated, and fault symptoms can be revealed and summarized for fault detection and diagnosis. This paper focus on reviewing dynamics based models and their applications.

In recent years, many dynamics based models have been developed and some review papers have been published. Wang et al. [26] reviewed non-linear mathematical models of gear systems and the corresponding solving methods. Parey and Tandon [27] reviewed spur gear dynamic models and touched the following gear faults: flank deviation, and wear and spalling. Bartelmus [28] gave a review of achievements in mathematical modeling and computer simulations for supporting fault detection in gearbox systems. Lei et al. [29] reviewed condition monitoring techniques for fault diagnosis of planetary gearboxes. But their main focus is not dynamics based gear modeling even though 11 references are described to mathematically simulate gear faults. Li et al. [30] reviewed the influence of planetary gear parameters on the dynamic characteristics of planetary gearboxes. The following parameters were considered: mesh phase difference, tooth profile modification, mounting errors, tooth profile modification and mesh stiffness. Cooley and Parker [31] gave a literature review on planetary gear dynamics and vibration mainly including mathematical models, vibration mode properties, dynamic response predictions, the effects of elastic compliance, and gyroscopic effects. Ma et al. [32] reviewed dynamics of cracked gear systems including crack propagation path modeling, time varying mesh stiffness evaluation and dynamics models for vibration analysis of cracked gear systems. Among these six review papers, only Refs. [27,32] elaborated on gear damage modeling methods. In Ref. [27], the modeling of flank deviation, wear and spalling were reviewed. However, this review paper was published in 2003. To date, many new research findings on gear damage modeling have been published. Ref. [32] gave a detailed literature review on gear crack modeling. But other types of gear faults are not touched. In addition, the above review papers all focus on gearbox modeling without stressing on techniques for fault diagnostics. Therefore, it is useful to make a systematic review on dynamics based gear damage modeling and its applications in gearbox fault detection and diagnosis. This review covers both fixed-axis and planetary gearboxes.

Dynamics based fault diagnostics relies on the analysis of simulated dynamic responses of a gearbox system to develop fault diagnosis techniques. First, a dynamic model is required to simulate a gearbox transmission system. In the modeling, some simplifications are generally applied to focus on key components and/or emphasize fault symptoms. Then, system responses are generated in gearbox healthy condition and faulty condition, respectively. Fault diagnosis techniques can be developed by analyzing gearbox responses. Meanwhile, a very important procedure is to conduct model validations to validate the dynamic models and the fault diagnosis techniques. The validation can be done by using an experimental system or comparing with other models. The overall process is summarized and given in Fig. 4. This process is a summary of the research methodology in the area of dynamics based gearbox fault diagnostics. A research paper is not necessary to have all the steps. It may only focus on gearbox modeling or system response analysis or model validation.

This paper aims to review dynamics based fault diagnosis techniques on all types of gear faults including gear tooth cracking, tooth pitting/spalling, wear, tooth tip chipping, manufacturing errors, misalignment, eccentricity and so on. There are many components in a gearbox, such as gears, bearings, casing, shafts, and couplings. A fault may occur on any of these components. This paper only reviews gear faults. The faults on other components of a gearbox are not considered. Both fixed-axis and planetary gearboxes are covered in this review. The remaining part of this paper is organized as follows. Section 2 gives a description of gearbox dynamic modeling techniques. Section 3 reviews gear mesh stiffness evaluation methods. Section 4 reviews gearbox damage modeling techniques and fault diagnosis techniques developed based on simulation. Section 5 reviews the methods for modeling the effect of transmission path. Section 6 reviews validation methods for mesh stiffness evaluation, dynamic modeling and fault diagnosis techniques, respectively. Section 7 gives a summary and some research prospects.

2. Gearbox dynamic modeling

Dynamic modeling utilizes physical laws, such as equilibrium, conservation of energy, and the Newton's laws of motion, to simulate gearbox system responses. The simulation does not have environmental noise interference. It can help understand gearbox dynamics, fault symptoms and fault generation mechanisms. In addition, the money cost to do dynamic simulation is negligible comparing with that for doing experiments. However, gearbox transmissions are very complicated and it is hard to model all details of a transmission. Researchers generally simplify a real system into a simplified discrete model while retaining all of the important and relevant features [1]. Computation burden of a simulation is another reason for the simplification of modeling. An event that may occur instantaneously in the real world may actually take hours or days to mimic in a simulated environment.

Lumped parameter modeling (LPM) and finite element modeling (FMM) are two commonly used techniques to model gear trains. A lumped parameter model is one in which the components are considered to be solid with the masses concentrated at a set of points [33]. A finite element model discretizes a physical model into disjoint components of simple geometry called finite elements and its system response is obtained by connecting or assembling the collection of all elements [33]. It is hard to simply tell which method is better. As stated in Ref. [34]: “Both methods are equally accurate if proper attention is paid to defining the boundary conditions and the degree of discretization, which may be different for the two methods. Solution costs will differ depending on the discretization characteristics of the LPM and various FMM derivations and, of course, on the efficiency of the programmer.”

2.1. Lumped parameter modeling

Many lumped parameter models were reviewed in [26,27,30–32] for fixed-axis and planetary gearbox transmission systems. We will not repeat reviewing these lumped parameter models again. Instead, two examples are given for a better understanding. One is for a fixed axis gearbox and the other one is for a planetary gear set.

Fig. 5 gives a model for a one-stage fixed axis gearbox. This model was originally proposed by Bartelmus [35]. Later, it was used for dynamic analysis of gears with tooth damages [36–39]. This model has 8 degrees of freedom: driving motor rotation, driven motor rotation, pinion rotation, gear rotation, x- and y-direction translational motions of the pinion and the gear,

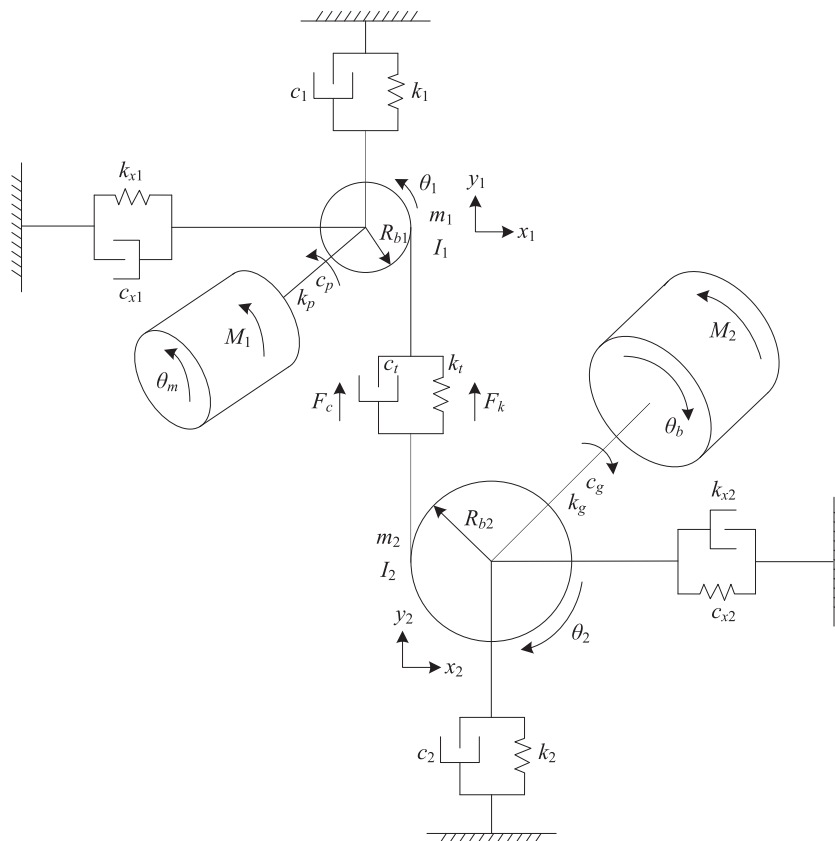


Fig. 5. Dynamic modeling of a one-stage gearbox [39].

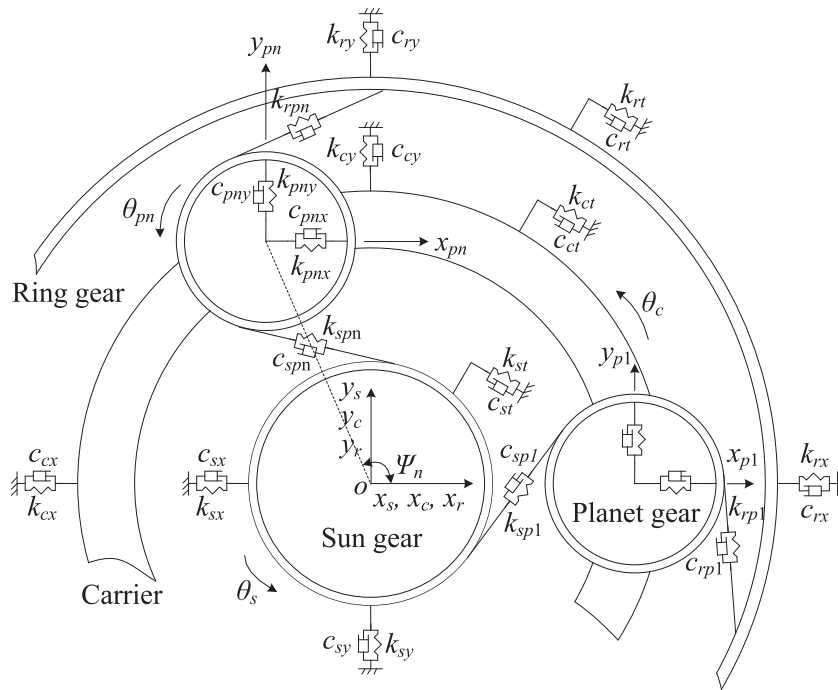


Fig. 6. Dynamic modeling of a planetary gear set [41].

respectively. Each gear is considered as a rigid plate with a central hole. Gear mesh interface and each bearing are modeled using a spring-damper system, respectively. The motor and gear shaft are coupled using a flexible coupling. This model ignored transmission errors, the frictions between gear teeth, shaft deflection, gearbox housing and other practical phenomena such as backlash. To be noticed that the x- and y-direction vibrations of gears are uncoupled due to the ignorance of the friction. Consequently, the vibration in the x-direction is free response. For free vibration, when system is stable, the vibration will disappear due to the inherent damping. Later, this model was improved by adding gear mesh friction [8,40].

Fig. 6 shows a typical 2-D lumped parameter modeling of a planetary gear set [41]. Each gear has three degrees of freedom: angular rotation and transverse motions in the x- and y-directions. Gear mesh interface and each bearing are modeled as a spring-damper system, respectively. Other practical phenomena such as gear transmission error, backlash, tooth friction, gear shaft deflection and gearbox housing are not considered. In this model, a rotating coordinate system is used to consider the inertia effect caused by the rotation of the carrier. All the translational motions are measured with respect to a rotating frame of reference fixed to the carrier. The planet deflections are described in the horizontal and vertical coordinates. A similar model is available in Ref. [42] where the planet deflections are presented in the radial and tangential coordinates. Meanwhile, a similar model but with a fixed coordinate system is given in Ref. [43].

In these two examples, gear damage is not involved in the lumped parameter modeling as a detailed review on gear damage modeling is given in Section 4.

2.2. Finite element modeling

Finite element modeling (FEM) is flexible to model any shaped gear and gear fault. But, it is sensitive to contact tolerances, mesh density and the types of finite elements selected [44]. As the mesh density increases, the numerical accuracy is improved, while the computational cost goes up [45]. To save the computation cost, Parker et al. [46] proposed a combined element/contact mechanics model to investigate the non-linear dynamic response of a spur gear pair. Later, this model was extended to investigate the dynamic response of a planetary gear system [47]. This finite element/contact mechanics approach did not require a highly refined mesh at the contacting tooth surfaces. In addition, the time-varying mesh stiffness and mesh contact forces were evaluated internally at each time step. The model reported in Ref. [47] was used to study the quasi-static loads [48] and the root stresses [49] in planetary gears, and the effect of manufacturing errors [50] and wear [51] on planetary gear dynamics. This element/contact mechanics model has been adopted in commercial software: Calyx [52]. Lin et al. [53] proposed a finite element method for 3D dynamic contact/impact problems of gears. This method is based on the derivation of the effective flexibility matrix equation, which is very efficient in computation for gear drives as only very small contact region is involved in tooth meshing at any time instant. Howard et al. [40,54–57] deeply investigated finite

element torsional models including gear mesh stiffness evaluation, crack and spalling modeling and gear dynamic characteristics. More finite element modeling details regarding to specific gear damages will be reviewed in Section 4.

3. Gear mesh stiffness evaluation

Gear mesh stiffness is one of the main internal excitations of gear dynamics [58]. It is time varying caused by the change of tooth contact number and contact position [59]. Section 3 describes two gear modeling techniques: lumped parameter modeling and finite element modeling. Lumped parameter modeling requires the mesh stiffness to be evaluated separately and considers the mesh stiffness as its model parameter, while finite element modeling can implicitly evaluate the mesh stiffness by the model itself. There are two ways to incorporate gear mesh stiffness in lumped parameter models. First, the mesh stiffness is pre-estimated and considered it as an input of the lumped parameter models [41,60–63]. Second, the mesh stiffness is evaluated at each time step as the gears rotate [46,50,64–70]. Both ways require the mesh stiffness to be evaluated explicitly. Therefore, the mesh stiffness evaluation methods to be reviewed in this section are mainly for the lumped parameter modeling.

Four methods are commonly applied to evaluate time-varying gear mesh stiffness for gear fault diagnosis: square waveform method, potential energy method, finite element method, and experimental method. In the following part, these four types of methods are reviewed one by one.

3.1. Square waveform method

For a healthy gear pair running constantly, gear mesh stiffness is a periodic function. Some researchers used a square waveform as shown in Fig. 7 to approximate gear mesh stiffness [61,71]. The period of a square waveform is called mesh period which equals to the time duration for one revolution divided by the number of teeth. The square waveform can reflect the change of tooth contact number but ignore the change of tooth contact position. The magnitudes of the square waveform are basically estimated based on personal experience. As we know, a square waveform can be approximated using Fourier Series. A few researchers used the first several terms of Fourier Series to represent the time-varying mesh stiffness [72,73].

When a fault occurs on a gear, gear mesh stiffness reduces. Most researchers expressed the mesh stiffness of a pair of gears with damage as follows:

$$k(t) = k_o(t) - \Delta k(t) \quad (1)$$

where $k(t)$ is the mesh stiffness of gears with damage, $k_o(t)$ represents the mesh stiffness of gears without damage and $\Delta k(t)$ denotes mesh stiffness reduction caused by gear damage. The time duration and magnitudes of $\Delta k(t)$ are determined subjectively based on the understanding of fault type and severity.

In Ref. [74], the mesh stiffness of gears with wear was obtained by adding amplitude and frequency modulations to the mesh stiffness curve of a perfect gear pair.

Table 1 lists the applications of the square waveform method to approximate gear mesh stiffness in healthy and damaged conditions. For a planetary gear set, the mesh phasing relationship [75] should be incorporated in the mesh stiffness evaluation. More details about the incorporation of mesh phasing will be given in Section 3.2.

The square waveform method is easy to use. However, the determination of mesh stiffness amplitudes and the reduction of the mesh stiffness caused by a fault are subjective without confirmation of physical parameters of gears.

3.2. Potential energy method

In the potential energy method, the gear tooth is assumed as a non-uniform cantilever beam and the beam theory is applied to evaluate gear mesh stiffness. This method was first proposed by Yang and Lin [100]. They considered the total energy stored in a pair of gears as the summation of Hertzian contact energy, bending energy and axial compressive energy corresponds to Hertzian contact stiffness k_h , bending stiffness k_b and axial compressive stiffness k_a , respectively. Later, Tian

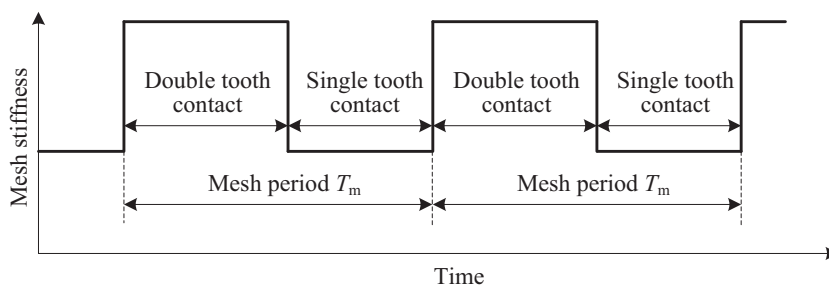


Fig. 7. Square waveform to approximate gear mesh stiffness.

Table 1
Applications of the square waveform method.

Type of gear train	Fault type	References
Fixed-axis	Healthy	[76–81]
	Tooth cracking	[82,83]
	Tooth pitting	[84]
	Tooth wear	[84,74]
	Broken tooth	[85]
	Tooth profile error	[82]
	Local fault	[86,73,87,88]
	Distributed fault	[89]
Planetary gears	Healthy	[71,90–93]
	Tooth cracking	[61,94,95]
	Tooth pitting	[61,96]
	Manufacturing error	[43,97,98]
	Tooth profile error	[99]

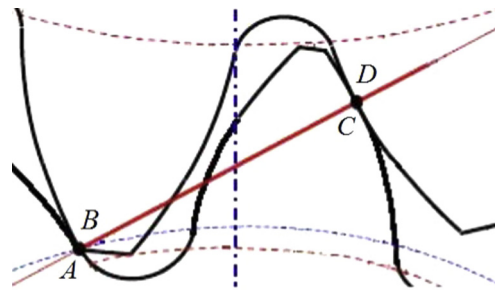


Fig. 8. Double tooth pairs in meshing.

[36] added another energy term called the shear energy corresponding to the shear stiffness k_s . The total effective mesh stiffness of a pair of gears is calculated as follows [36]:

$$k_t = \sum_{i=1}^m \frac{1}{\frac{1}{k_{h,i}} + \frac{1}{k_{b1,i}} + \frac{1}{k_{s1,i}} + \frac{1}{k_{a1,i}} + \frac{1}{k_{b2,i}} + \frac{1}{k_{s2,i}} + \frac{1}{k_{a2,i}}} \quad (2)$$

where i represents the i th pair of meshing teeth, the subscripts 1 and 2 denote the driving gear and the driven gear, respectively.

Some researchers also incorporated the gear body effect (also called the effect of tooth fillet foundation deflection) in the mesh stiffness evaluation [5,60,101–103]. In almost all these papers, the fillet foundation effect was evaluated using Eq. (17) of Ref. [104]. This equation was derived assuming linear and constant stress variations at the root circle. This assumption confined this equation to be usable only for large gears [104]. In addition, this equation was derived for a perfect gear with a single tooth in meshing. If more than two pairs of tooth contacts take place simultaneously, the accuracy of this equation is doubtful. In addition, if the gear body has damage, like crack, this equation may not be valid.

For a pair of spur gears without tooth profile modification, the alternation of one pair and two pairs of teeth in contact is observed. Fig. 8 gives the scenario of two pairs of teeth in contact. Point A of gear 1 is touching point B of gear 2 meanwhile point C of gear 1 is touching point D of gear 2. If we know these four contact points, the mesh stiffness of a pair of gears can be evaluated easily using the potential energy method. However, with the rotation of gears, the location of these four points changes. One challenge in using the potential energy method is how to get the location of these four points at any rotation angle of gears. This problem was solved by deriving the mesh stiffness equations of a pair of gears as a function of gear rotation angle. In Refs. [36,37,58,59,105,103], the Hertzian contact stiffness, bending stiffness, shear stiffness and axial compressive stiffness are all derived as a function of gear rotation angle. Correspondingly, the total effective mesh stiffness of a pair of gears is also a function of gear rotation angle through Eq. (2). Users can use these equations directly to evaluate gear mesh stiffness even though they are not familiar with beam and gear meshing theories.

For a gear transmission with multiple gear pairs meshing simultaneously (like a planetary gear set or an idler gear system), the mesh stiffness of each gear pair is evaluated separately, then, by incorporating the mesh phasing relationships, the mesh stiffness of the whole gear transmission can be synchronized. Fig. 9 illustrates the procedures to evaluate the mesh stiffness of a basic planetary gear set. In a basic planetary gear set, there are multiple sun-planet gear pairs and ring-planet gear pairs meshing simultaneously. The mesh stiffness of a pair of sun-planet gear and a pair of ring-planet gear is evaluated separately using the methods develop for fixed-axis gear pairs. Then, consider the mesh phasing relationships

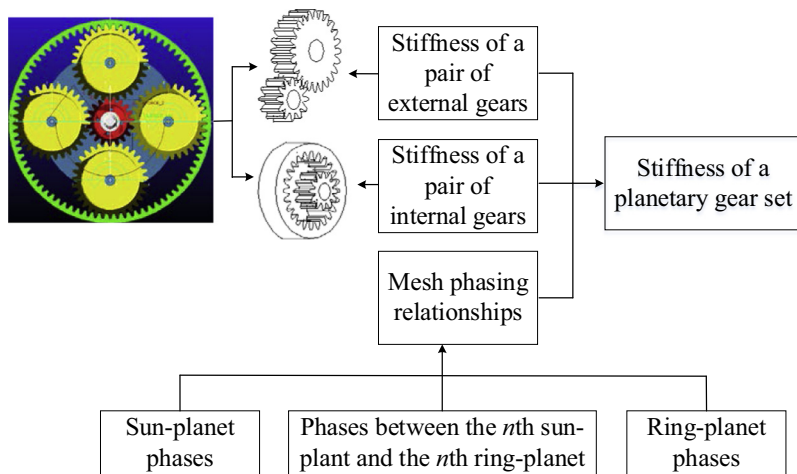


Fig. 9. Mesh stiffness evaluation of a basic planetary gear set [106].

Table 2

Applications of the potential energy method.

Type of gear train	Fault type	References
Fixed-axis	Healthy	[100,107,108],
	Tooth cracking	[36–38,103,109–111,62,102,112–114,8,115–124]
	Tooth pitting/spalling	[5,39,58,125–128,120]
	Tooth wear	[129]
	Broken tooth	[5,36]
	Tooth chipping	[36]
	Tooth profile error	[130]
	Misalignment	[131,132]
Planetary gears	Healthy	[105,133,134]
	Tooth cracking	[59,41,106,135–137,60,138–142]
	Tooth pitting	[142]

[36,41], the mesh stiffness of the whole planet gear set can be obtained. The mesh phasing relationships include sun-planet phases, ring-planet phases and the phases between the n th sun-planet and the n th ring-planet.

If damages occur on a gear tooth, the effective tooth thickness and/or gear tooth contact length and/or effective tooth length are different comparing with that of perfect teeth. By counting these changes, gear mesh stiffness reduces. The location and amount of the mesh stiffness reduction depends on fault profile, fault location and fault severity.

Table 2 gives a list of the applications of the potential energy method for evaluating the time-varying mesh stiffness of gears in healthy and damaged conditions.

In the potential energy method, the contribution of individual stiffness component, like bending stiffness, shear stiffness, axial compressive stiffness and Hertzian contact stiffness, can be analyzed separately. The mesh stiffness is directly related to fault type, location and severity. But, gear meshing theory is required in mesh stiffness equation derivation unless the mesh stiffness equations are expressed as a function of gear rotation angle as did in Refs. [37,58,59].

3.3. Finite element method

As described in the beginning of Section 2.2, finite element modeling based gearbox dynamic analysis does not require the gear mesh stiffness evaluated explicitly. A few researchers used the finite element method to evaluate time-varying mesh stiffness of gears mainly for two purposes: (a) substitute the stiffness results into a lumped parameter dynamic model to investigate gear dynamics and (b) validate the mesh stiffness result obtained using other methods. In this section, we mainly focus on the research for the first purpose. Studies for the second purpose will be further discussed in Section 6.1. Ma et al. [32] classified the finite element method into three sub-categories: finite element models with contact elements, finite element models by applying contact load and combined analytical-finite element approach. Cooley et al. [143] described two approaches for spur gear tooth mesh stiffness evaluation: the average slope approach and the local slope approach. The average slope approach is preferred for gear static analysis while the local slope approach is good for gear dynamic analysis. Liang et al. [58] used linear finite element analysis to save computation cost in gear mesh stiffness evaluation of gears with tooth pitting. Howard et al. [40,54–57] developed finite element models to evaluate gear torsional mesh stiffness. Ambarisha and

Table 3
Applications of the finite element method.

Type of gear train	Fault type	References
Fixed-axis	Healthy	[53,54,56,67,68,143,145,146–148]
	Tooth cracking	[40,55,62,149–152]
	Tooth pitting/spalling	[55,58,150,152]
	Tooth broken	[153,154]
	Tooth profile error	[155]
	Misalignment	[156]
Planetary gears	Healthy	[144]
	Tooth cracking	[57]
	Tooth wedging	[157]

Parker [144] used a combined element/contact mechanics model to calculate the mesh stiffness of a planetary gear set. This finite element/contact mechanics model does not require a highly refined mesh at the contacting tooth surfaces. The sun-planet mesh stiffness and ring-planet mesh stiffness is uncoupled by fixing the planet gear and assuming rigid gear bearings. The gear mesh phasing relationship is added afterward according to Ref. [75]. Eritenel and Parker [68] proposed a three-dimensional nonlinear dynamic model and compared the difference between static mesh stiffness and dynamic mesh stiffness. Pandya and Parey [62] investigated the effect of crack propagation paths on spur gear mesh stiffness using finite element method. A list of applications of the finite element method for evaluating mesh stiffness of gears is presented in Table 3.

Finite element method is capable of evaluating the gear mesh stiffness for any type of gear profile and gear fault but the result is sensitive to contact tolerances, mesh density and the types of finite elements selected [44]. An accurate mesh stiffness evaluation requires the users to know both gear meshing theory and finite element modeling theory.

3.4. Experimental method

Li et al. [158] proposed to measure gear tooth dynamic deformation using dynamic speckle photography. Amarnath et al. [159,160] experimentally measured gear mesh stiffness using modal analysis in conjunction with a theoretical model for surface wear assessment. Pandya and Parey [161] experimentally measured the mesh stiffness of a pair of spur gears with crack using the photoelasticity technique. Raghuwanshi and Parey [162] applied the photoelasticity technique to measure the stress intensity factor of a cracked gear tooth. Then, the stress intensity factor is used to evaluate the gear mesh stiffness. Raghuwanshi and Parey [163] proposed an experimental technique based on strain gauge to measure the gear mesh stiffness of a spur gear pair with tooth crack. Munro et al. [164] proposed an experimental method to measure gear tooth stiffness based on the properties of transmission error plots for mean and alternating components over a range of tooth loads. Wang and Morrish [165] calculated gear mesh stiffness using test box deflection measurements and theoretical considerations. Chi et al. [166] experimentally measured static individual nylon gear torsional mesh stiffness and investigated the relationship between the static torsional mesh stiffness and static transmission error. Park et al. [167] proposed experimental procedures to estimate gear mesh stiffness of rotating gears under faulty states. Li et al. [158] and Park et al. [167] claimed their method was usable and effective when the machine was rotating, but they did not test their method in very high speed. All other experimental methods were only tested under static conditions.

4. Gearbox damage modeling and fault diagnosis techniques

In this section, a detailed literature review gives to gear damage modeling, dynamic responses and fault diagnosis techniques. The literature review is classified based on gear fault types.

4.1. Gear tooth cracking

Ma et al. [32] did a literature review on gear tooth crack modeling. We will not repeat the review but a short description is given. In addition, some papers not covered in Ref. [32] are added.

Based on finite element modeling studies, Lewicki [168] found that gear crack propagation path was affected by many factors, such as rim and web thicknesses, initial crack location and backup ratio (rim thickness divided by tooth height). Fig. 10 gives an example showing the effect of initial crack location on crack propagation path. More details about the effect of other factors on crack propagation path is available in Ref. [168]. Belsak and Flasker [169] concluded that gear crack propagation paths were smooth, continuous, and in most cases, rather straight with only a slight curvature. Kramberger et al. [170] indicated that crack mostly initiated at the point of the maximum principal stress in the tensile side of a gear tooth.

In dynamic modeling, gear tooth crack paths are usually simplified to a straight line [37,59] or a slight curve [62,171] initiating in the tooth root as shown in Fig. 11. Most research papers assumed tooth crack going through the whole tooth width with a constant crack depth. Chen and Shao [101] first proposed a crack model propagating along both tooth width and crack

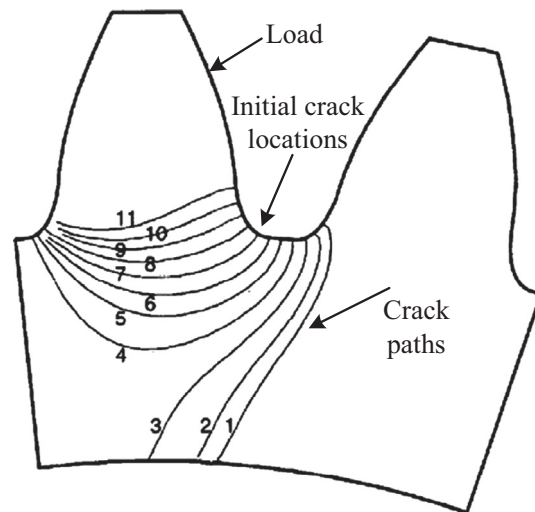


Fig. 10. Effect of initial crack location on crack propagation path [168].

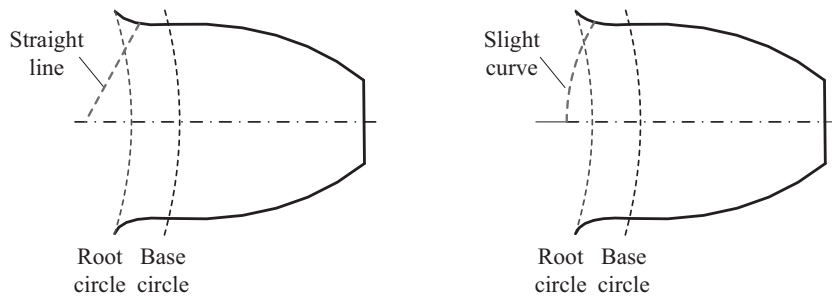


Fig. 11. Simplified models for gear tooth crack path.

depth. The crack effect can be simulated using the lumped parameter modeling or the finite element modeling [32]. In the lumped parameter modeling, gear crack fault is mostly reflected in the time-varying mesh stiffness. Tooth crack causes gear mesh stiffness reduction leading to abnormal vibration of gears. Gear mesh stiffness can be evaluated using the square wave-form method, the potential energy method, the finite element method or the experimental method as described in Section 3. In the finite element modeling, mesh stiffness variation excitation and tooth contact are intrinsically considered [144].

In this paragraph, some research papers not covered in Ref. [32] are reviewed. Cheng et al. [94] developed a crack level estimation methodology for a planetary gearbox by integrating a lumped parameter model for simulation signal generation, a statistic algorithm for feature selection and a grey relational analysis (GRA) algorithm for damage level estimation. The crack level estimation methodology was validated using an experimental test rig. Lei et al. [72] developed a nonlinear dynamic model to simulate torsional vibration of a planetary gear train with a cracked carrier plate. Time and frequency domain vibration signals were investigated and fault features were extracted. In [72,94], the mesh stiffness reduction caused by tooth crack is approximated using the square waveform method. Tian et al. [38] used a 6 degrees of freedom (DOF) lumped parameter model to simulate a one-stage gearbox with spur gears and one tooth crack. Statistical indicators and the discrete wavelet transform technique were investigated to identify effective health indicators to reflect the crack propagation level. Liang and Zuo [137] utilized an 18DOF lumped parameter model to investigate vibration properties of a planetary gear set with a cracked tooth in a planet gear in the time and frequency domains. The findings were validated to some extent by comparing with an experimental test rig. Li et al. [140] developed crack detection techniques for a multistage planetary gear set using dynamics based simulation and experimental investigation. Liu et al. [134] proposed a comprehensive vibration signal model for a planetary gear set based on a two-dimensional lumped-parameter dynamic model. Transmission path effects were modeled as two parts: the part inside the gearbox to the casing and the other part along the casing to the transducer position. Hu et al. [122] used wavelet analysis on dynamic simulated signals of gears for crack detection. In Refs. [38,122,134,137,140], the tooth crack was modeled in a straight line shape and its effect was reflected by gear mesh stiffness which was evaluated using the potential energy method. Cui et al. [111] investigated the effect of tooth crack inclination angle on the mesh stiffness and vibration response of a pair of gears based on the universal equation of gear profile. Cai

et al. [151] studied the stress intensity factors for a crack in an involute spur gear tooth to predict the fatigue crack damage. Mohammed and Rantatalo [117] modeled crack propagating in both the depth and the length directions simultaneously, simulated gear vibration signals using a 6DOF lumped parameter model and investigated gear tooth crack detection by using nature frequencies and performing time-frequency analysis. Endo et al. [172] proposed a differential diagnostic technique based on static and dynamic simulation models to diagnose two types of localized gear tooth faults: a spall and a crack in the gear tooth fillet region. Ma et al. [119] introduced time varying mesh stiffness into a finite element model of a gear rotor system and investigated the combined effects of tooth crack and gear profile shift under each of the positive, negative and zero gear transmission conditions on system vibration responses. Chen et al. [114] modeled non-uniformly distributed tooth root crack along tooth width using a lumped parameter model for a pair of fixed-axis gears and investigated its effect on gearbox dynamic responses. Liang et al. [106] proposed a signal decomposition technique to enlarge gearbox fault symptoms. If a planet gear has N teeth, the transducer perceived vibration signal can be decomposed into N sub-signals. Each sub-signal corresponds to one tooth of a planet gear. Examining the differences of these N sub-signals, the health differences of these N teeth can be measured. Saxena et al. [121] studied the effect of mesh stiffness of healthy and cracked gear teeth on modal and frequency response characteristics of a geared rotor system. Yin et al. [173] proposed a nonlinear dynamic model to describe gear tooth crack growth under variable-amplitude loading. Walha et al. [174] investigated dynamic response of a two-stage gear system damaged by tooth crack and teeth scaling, respectively.

4.2. Gear tooth pitting/spalling

Tooth pitting is a common failure mode of gears. When several pits join, a larger pit (or spall) is formed. According to American Society for Metals (ASM) handbook [175], the main causes of tooth pitting/spalling are summarized as follows:

- Subsurface cracks caused by figure or inclusions in gear materials
- Metal-to-metal contact of asperities or defects due to low lubricant film thickness
- Foreign particle contamination of lubricant

Tan et al. [176] experimentally measured tooth pitting growth under different speed and load conditions. The lubricant oil (SAE 20W-50) they used does not have anti-wear properties to accelerate tooth pitting propagation. Fig. 12 gives an example of experimental test results of tooth pitting propagation under the working condition of 73 Nm and 745 rpm. The tooth pitted area is 6.3%, 27.8% and 41.7% after 48.5 h, 240.5 h and 402.5 h of operating, respectively. The pitted area percentage is the ratio of pitted area to the surface area of the meshing side of a tooth. Their experimental results show pitting propagates to neighboring teeth and finally spreads to other gear teeth with prolonged operation time.

Chaari et al. [61] compared healthy planetary gear dynamic response and the response of planetary gears containing tooth defects (tooth pitting and cracking) in both the time and frequency domains and in the joint time frequency domains using the Wigner-Ville distribution. The faults were approximated by giving a gear mesh stiffness reduction using the square waveform method. Chaari et al. [5], Zhe et al. [96] and Abouel-seoud et al. [142] modeled a single gear tooth pit in rectangular shape as shown in Fig. 13(a). Chaari et al. [5] investigated a single tooth pit or tooth breakage effect on gear mesh stiffness and gear vibration frequency spectrum. Zhe et al. [96] studied a single tooth pit effect on the vibration of a planetary gear set and used the grey relational analysis to estimate pitting damage levels. Abouel-seoud et al. [142] presented a single rectangular tooth pit effect on the mesh stiffness of a wind turbine gearbox and analyzed vibration signal fault signatures in the time and frequency domain. Ma et al. [126] modeled a rectangular spalling and investigated the influence of spalling width, spalling length and spalling location on mesh stiffness, respectively. Saxena et al. [127] modeled a single spall in the rectangular, circular or V-shaped, respectively and investigated the effect of spall shape, size and location on gear mesh stiffness. Liang et al. [58] modeled tooth pitting propagation on a single tooth by increasing the number of circular pits as shown in Fig. 14. The mesh stiffness equations are derived using the potential energy method and expressed as a function of gear rotation angle, which is easy to use. Later, Liang et al. [39] modeled tooth pitting propagation to neighboring teeth

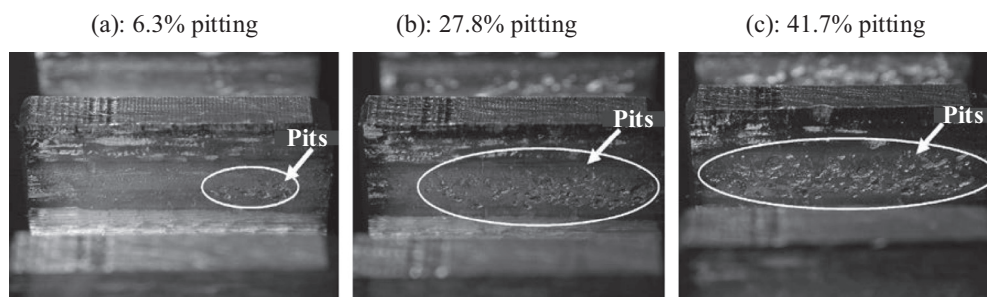


Fig. 12. Pitting propagation under the operation condition of 73 Nm and 745 rpm [176].

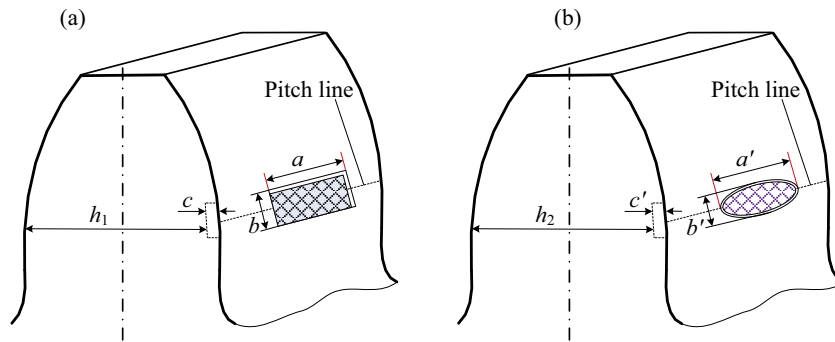


Fig. 13. A single tooth pit modeling [58].

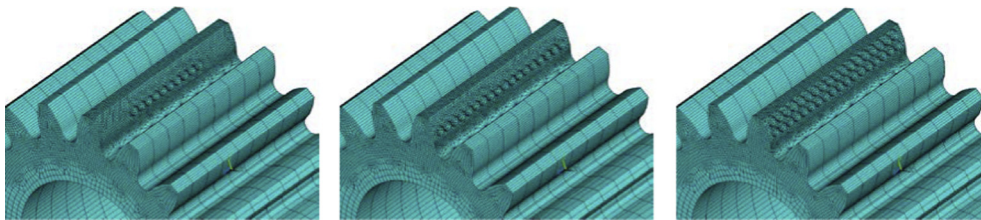


Fig. 14. Pitting growth modeling (slight to moderate to severe – from left to right) [58].

and analyzed fault symptoms for pitting damage detection. In addition, statistical features that are insensitive to gear mesh damping and environmental noise are recommended. In Refs. [5,39,58,96,126,127,142], gear fault is reflected by gear mesh stiffness reduction which is evaluated using the potential energy method. Li and Kahraman [177] proposed a physics-based model to predict the micro-pitting behavior on contact surfaces of spur gears operating under the mixed lubrication condition. Jiang et al. [128] modeled the spalling defect as a rectangular indentation near the pitch line, and parallel to the contact line on one of the tooth. Then, they investigated the dynamic characteristics of a pair of helical gears under sliding friction with spalling effect using a 6DOF lumped parameter model. Ma et al. [70] investigated fault mechanism of a spur gear pair with spalling defect. Later, Ma and Chen [120] used a four-degree of freedom gear system with local crack or spalling fault to investigate the failure mechanism and characteristics. The two types of failures were compared using time history vibration, phase contrail, Poincaré section, spectrum analysis and fractal dimension. Experimental results agreed with the theoretical results to some extent. Rincon et al. [152] used the finite element methods to analyze a single elliptical pit (see Fig. 13(b)) effect on mesh stiffness. Parey et al. [178] investigated the dynamic responses of a 6 DOF lumped parameter model and concluded that empirical mode decomposition pre-processed kurtosis and crest factor give early detection of pitting as compared to raw signal. The gear tooth profile error, eccentricities and defect were incorporated in the dynamic model by modifying the dynamic mesh force profile. Chaari et al. [179] investigated the effect of gear tooth pitting and crack on dynamic response of a planetary gear set, respectively.

4.3. Tooth wear

According to ASM handbook [175], “Gearsets are susceptible to wear caused by adhesion, abrasion, and polishing. Adhesive wear is classified as “mild” if it is confined to the oxide layers of the gear tooth surfaces. If, however, the oxide layers are disrupted and bare metal is exposed, the transition to severe adhesive wear usually occurs. Scuffing is defined as localized damage caused by solid-phase welding between sliding surfaces. Abrasive wear on gear teeth is usually caused by contamination of the lubricant by hard, sharp-edged particles. If the extreme-pressure antiscuff additives in the lubricant are too chemically reactive, they may cause polishing of the gear tooth surfaces until they attain a bright mirror finish. Although the polished gear teeth may look good, polishing wear is undesirable because it generally reduces gear accuracy by wearing the tooth profiles away from their ideal form.”

Choy et al. [84] simulated the effects of tooth surface pitting, wear and partial tooth fracture on the vibration signal of a gear transmission system. Gear faults were simulated using mesh stiffness change based on the square waveform method. The Wigner-Ville Distribution method was demonstrated to be effective in detecting and locating tooth wear and pitting damage. Experiments results obtained from a gear test rig at NASA Lewis Research Center were used to verify their research findings. Flodin and Andersson [180] proposed a numerical model to simulate mild wear in spur gears based on a generalized Archard's wear equation [181]. Later, they developed a computer model to simulate the wear behavior in helical gears [182].

Bajpai et al. [183] proposed a surface wear prediction method for spur and helical gears. Their methodology adopted a finite elements-based gear contact mechanics model in conjunction with the Archard's wear equation to predict wear of contacting tooth surfaces. Liu et al. [184] modified a gear mesh stiffness evaluation model reported in Ref. [185] to simulate gear mesh contact for a spur gear with and without tooth damage, and study the influence of gear design parameters on gear tooth damage detection sensitivity. Three gears faults were investigated including tooth pitting, root crack and wear. Yesilyurt et al. [186] developed a model analysis technique to assess wear damage of spur gears and a vibration-based experimental procedure was designed to validate their method's accuracy. Yuksel and Kahraman [51] developed a deformable-body dynamic model to study the influence of surface wear on the tooth loads of a typical automotive automatic transmission planetary gear set. A 2DOF rotational model was used in Ref. [73] for a pair of spur gears to predict sideband amplitude in the presence of distributed gear faults such as non-uniform wear and pitting. Gear faults were embedded by adding an excitation function to the differential equations of motion. Feng et al. [129] evaluated gear mesh stiffness with tooth wear using the potential energy method. Ming [187] studied friction and wear behaviors of gear steel under coupling of rolling and sliding. Cui and Cai [188] developed a gear-shaft-bearing dynamic model to investigate the dynamic behavior of a gear system with breathing crack and tooth wear fault. Tooth wear was reflected by modifying gear mesh stiffness shape. In Ref. [189], two different dynamic models namely a finite elements-based deformable-body model and a simplified discrete model were combined with the Archard's wear model to study the interaction between surface wear and dynamic response of a pair of gears. Kahraman and Ding [190] combined a torsional dynamic model and the Archard's wear model to investigate the interactions between the surface wear and the dynamic behavior of planetary gears within both linear and non-linear response regimes. Liu et al. [74] proposed a gear wear prediction methodology by combining a translational-rotational-coupled nonlinear dynamic model with a quasi-static wear model to investigate the coupling between tooth surface wear and dynamic behaviors of a spur gear system.

4.4. Tooth tip chipping

Gear tooth tip chipping refers to a small piece of material breaking away from the tip of a tooth. According to [191], "Failures of this kind may be caused by deficiencies in the gear tooth, which results in a high stress concentration at a particular area. Sometimes flaws or minute grinding cracks will propagate under repeated stress cycling and a fracture will eventually develop. Foreign material passing through the gear mesh will also produce short-cycle failure of a small portion of a tooth. High residual stresses due to improper heat treatment can cause local fractures that do not originate in the tooth root section."

Tian et al. [36] modeled tooth chipping in a curved shape, evaluated chipping effect on gear mesh stiffness using the potential energy method, simulated gearbox vibration signals using a 6DOF lumped parameter model and investigated fault symptoms using simulated vibration signals. Wang et al. [192] used a lumped parameter model to investigate sideband frequency components around gear mesh frequency. Tooth chipping was modeled by adding periodic impulses to gear mesh stiffness profile. Results of experimental signal analysis matched the theoretical findings. Chaari et al. [5], Cheng and Hu [193], and Li et al. [194] modeled tooth chipping in straight line shape and the chipping effect was reflected in gear mesh stiffness. Chaari et al. [5] evaluated gear mesh stiffness using the potential energy method, simulated vibration signals for a one stage spur gear transmission using a lumped parameter model, and investigated chipping effect on gear mesh stiffness and vibration signal frequency spectrum. Cheng and Hu [193] utilized statistical features to quantify sun gear chipping severity of a planetary gear set using a lumped parameter model. Li et al. [194] investigated chipping effect on gear mesh stiffness using the potential energy method and vibration responses of a compound planetary gear set using a lumped parameter model. Research findings were validated by comparing with experimental results. Jiang and Liu [153] developed a three-dimensional analytical helical gear pair model to investigate the effect of tooth breakage with sliding friction considered on the dynamic responses of helical gears.

4.5. Manufacturing errors

Gear manufacturing errors are generated in the production process of each individual gear, such as profile errors and tooth spacing errors [31]. Cooley and Parker [31] reviewed the investigations on manufacturing errors for planetary gears. This section reviews investigations on manufacturing errors for fixed-axis gears and some papers not covered in Ref. [31] for planetary gears.

Chaari et al. [97] developed a plane model of a planetary gear train to investigate the effect of manufacturing errors (eccentricity or profile error) on the frequency response of gears. Velez and Maatar [64] proposed a unified three-dimensional lumped parameter model to analyze the influence of shape deviations and mounting errors on gear dynamics. In their model, "dynamic mesh stiffness and transmission error are some of the results of the procedure and are not introduced as excitation functions determined by a prior static analysis." Ottewill et al. [195] combined a dynamic model of an epicyclic gearbox with a simple space-phaser model of an induction motor to test the possibility of using motor current signals to identify tooth profile faults. Their results were validated by applying the synchronous averaging approach to experimentally recorded motor currents and angular displacements recorded from a shaft mounted encoder. Gear fault was modeled by modifying the static transmission error. MacLennan [155] proposed an analytical model to determine the influence of shape deviation on load distribution and mesh stiffness of spur gears. Gear contact forces were measured to validate

their results. Wang and Zhang [130] built a model for analyzing tooth shape deviations and assembly errors on helical gear mesh stiffness, loaded transmission error, tooth contact stress and tooth root stress. Jia et al. [82] built a dynamic model for multiple spur gear pairs and investigated the effect of tooth crack, pitch and profile errors on the angular velocity of gears. Pitch and profile errors were approximated using a combination of three harmonic terms. Koch and Vicuña [99] used dynamic and phenomenological vibration models for planet gear tooth flank error prediction of planetary gearboxes. Gear fault was reflected in gear mesh stiffness evaluated using the square waveform method. Time domain waveform and frequency spectrum were investigated and compared with experimental signals for fault detection. Walha et al. [196] investigated effects of manufacturing defects on the dynamic behavior for an helical two-stage gear system. Fakhfakh [197] modeled three types of gear manufacturing errors (eccentricity, tooth profile error and assembly defect) for a two-stage gear system and analyzed the effects of defects on gear system dynamic behavior.

4.6. Misalignment and eccentricity

Saxena et al. [132] investigated the effect of shaft misalignment and friction force on time varying mesh stiffness of a spur gear pair. Cao et al. [131] studied the effect of carrier misalignment error on the vibration properties of a planetary gear set. Gu and Velex [198] found that planet pin-hole errors would alter tooth load distributions and lead to tooth separations and shock for lightly damped spur gears using a non-linear dynamic model of planetary gears. Gu and Velex [65] developed a lumped parameter model to account for planet position errors and simulated their contribution to the quasi-static and dynamic load sharing amongst the planets. Gu and Velex [66] simulated the contributions of eccentricity errors to the quasi-static and dynamic behavior of planetary gears using a dynamic model. Yu et al. [199] used a dynamic model to investigate the dynamic coupling behavior of a cylindrical geared rotor system with local tooth profile errors and global mounting errors. Yassine et al. [83] used square waveform to approximate gear mesh stiffness and investigated dynamic responses of a two-stage straight bevel system with eccentricity defect, profile error and cracked tooth, respectively. Time domain waveform and frequency spectrum were analyzed for fault detection. Peng et al. [200] studied eccentricity effect on right-angle gear dynamics. Hbaieb et al. [201] studied influences of eccentricity, profile error and tooth pitting on the dynamic response of a planetary helical gear system, respectively. Abbes et al. [202] developed a dynamic model of a spur gear system using finite element method combined with elastic foundation theory. Then, eccentricity error and tooth crack effect on dynamics behavior was investigated. Walha et al. [203] investigated effects of eccentricity defect on the nonlinear dynamic behavior of an automotive clutch coupled with a helical two-stage gear system.

4.7. Other fault types

Litak and Friswell [87] examined the dynamics of gear systems with various faults in meshing stiffness. The faults were modeled by reducing gear mesh stiffness using the square waveform method. Bartelmus [86] showed how mathematical modeling and computer simulation help the interpretation of signals taken from real measurements. He investigated local gear fault and distributed gear fault of a one-stage gearbox, respectively. Bartelmus et al. [89] modeled gearbox dynamics under time-varying nonstationary load for distributed fault detection and diagnosis. The simulation results were found to be in agreement with experimental results. In [86,89], gear faults were modeled by adding error functions to the calculated mesh displacement along the line of action. Guo and Parker [157] used a 2D planetary gear dynamic model to reveal that tooth wedging elevated planet bearing forces dramatically and destroyed load sharing among the planets. Park et al. [95] developed a dynamic model based fault diagnosis technique of a planetary gear set by using the transmission error signal. Diehl and Tang [154] developed a dynamic gearbox model for a two-stage gearbox and used the harmonic wavelet transform analysis to identify fault type and fault severity. Their method was validated on a test bed. Han et al. [204] investigated steady state response of a geared rotor system with slant cracked shaft and time varying mesh stiffness. Forced response spectra were investigated for shaft crack detection in a geared rotor system. Su et al. [85] used square waveforms to approximate time-varying mesh stiffness and gear fault features were investigated in time and frequency domains. Cui et al. [205] studied nonlinear dynamics of gear system considering unbalance and loosening faults. Begg et al. [206] developed a finite element based dynamic model for a gearbox system. Statistical features were used to monitor the health condition of the gearbox system for the purpose of fault diagnosis and prediction. Rincon et al. [207] proposed an enhanced model of gear transmission dynamics with the effects of torque, friction and bearing clearance considered, which has the potential to be used for condition monitoring. Wang [88] proposed a gear dynamics fault (crack) diagnosis/prognosis model at the Aeronautical and Maritime Research Laboratory, Australia. A resonance model was adopted to adaptively update model parameters by minimizing the power of error between the modeled and measured signals. Karray et al. [208] studied dynamic response of a single stage bevel gear transmission system in the presence of local damage. Chaari et al. [209,210] investigated local damage effect on gearbox dynamics under non-stationary operations. Bartelmus and Grabski [211] developed a nonlinear dynamic model using Mathematica software and investigated the load susceptibility characteristics for a pair of gears with pitch error. Abbes et al. [212] studied effects of transmission error on the dynamic behavior of gearbox housing. Bartelmus and Zimroz [213] and Fakhfakh et al. [214] built mathematical models to simulate vibration signals of gears with localized and distributed faults, respectively.

5. Effect of transmission path

In real applications, transducers are usually installed on the housing of gearboxes or bearings to acquire vibration signals. The vibrations inside a gearbox go through different transmission path to reach a transducer. Some studies focus on simulated vibration analysis of individual gears (displacement, velocity and acceleration analysis) or one gear pair (dynamic force analysis). In these studies, the effect of transition path is not considered. But, analysis of individual gears or gear pairs provides us a way to look inside of a gearbox and helps us understand the characteristics of each vibration source. A few studies have considered and modeled the effect of transmission path to analyze the response of a whole gearbox. Modeling and analysis of the vibration of a whole gearbox can reveal the process of vibration generation, tell the best location for transducer installation and promote the development of effective fault diagnosis techniques. This section gives a literature review on transmission path modeling and analysis.

5.1. Fixed-axis gearbox

Lim and Singh [215–217] investigated the effect of vibration transmission path through rolling element bearings. They proposed a bearing stiffness formulation to consider the transmission path between gears and the housing. He [145,218] developed a source-path-receiver model of a single gear pair system to quantify the effect of sliding friction between gear teeth on the structure-borne whine noise. Omar et al. [219] developed an improved gearbox model by adding the modeling of gearbox foundation and the plate where accelerometers were mounted. Hambric et al. [220,221] developed a computational model to simulate the effects of rolling element and journal bearings on the vibration and sound transmission through gearboxes. Parker et al. [222,223] built a gear-bearing-housing system using a combination of mathematical modeling and finite element analysis to investigate the vibro-acoustic propagation of gear vibration. However, gear faults were not investigated in their study. Guo et al. [224,225] developed a hybrid model to study the transmission path characteristics of gear-shaft-bearing-housing. Ambarisha et al. [226] compared the performance of two gearbox housing models. The first model was represented by a full FE mesh and the second model was replaced by a reduced model with condensed stiffness and mass matrices.

5.2. Planetary gearbox

A planetary gear set has several sun-planet and ring-planet gear pairs meshing simultaneously. In addition, carrier rotation may cause the distances between vibration sources and a transducer time varying. Therefore, transmission path of a planetary gearbox is more complicated than that of a fixed-axis gearbox. Fig. 15 shows three possible transmission paths for a sun-planet meshing, (a) meshing point – planet gear – housing – transducer, (b) meshing point – planet gear – planet gear bearing – carrier – carrier bearing – housing – transducer and (c) meshing point – sun gear – sun gear shaft – sun gear shaft bearing – housing – transducer. Inalpolat and Kahraman [23] and Feng and Zuo [24] simulated transducer perceived vibration signals of a planetary gearbox based on amplitude modulation and frequency modulation characteristics of vibration signals. Inalpolat and Kahraman [98] proposed a dynamics based method to simulate transducer perceived acceleration signal $a(t)$ which was expressed as follows:

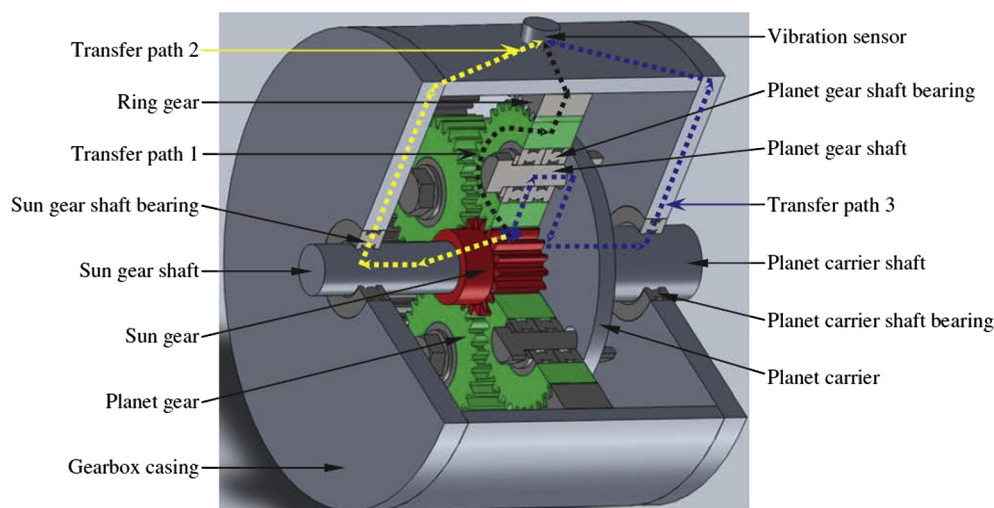


Fig. 15. Illustration of transmission path [24].

$$a(t) = \sum_{n=1}^N (C_s w_n(t) F_{spn}(t) + C_r w_n(t) F_{rpn}(t)) \quad (3)$$

where C_s and C_r are constants to express the influence weight of gear mesh forces, $w_n(t)$ denotes the Hanning function to represent the effect of transmission path, and $F_{spn}(t)$ and $F_{rpn}(t)$ means dynamic forces of the n th sun-planet mesh and the n th ring-planet mesh, respectively. In Refs. [23,24,98], the effect of transmission path was modeled using a Hanning function based on an assumption that as planet n approaches the transducer location, its influence increases and reaches its maximum when planet n is closest to the transducer location, then, its influence decreases to zero as the planet goes away from the transducer. Liang et al. [136,137] proposed to use the Hamming function to represent the effect of transmission path. Liang et al. [41] proposed a modified Hamming function to represent the effect of transmission path and considered the transducer perceived vibration as the weighted summation of the vibration of each planet gear:

$$a(t) = \sum_{n=1}^N e^{\alpha(\text{mod}(w_c t + \psi_n, 2\pi) - \pi)^2} H_m(t) a_n(t) \quad (4)$$

where w_c is carrier rotation speed, ψ_n denotes circumferential angle of the n th planet gear, $H_m(t)$ represents the Hamming function for the n th planet gear, $a_n(t)$ is the acceleration signal of the n th planet gear, and a is a weighting factor to control the bandwidth of the Hamming function. Liu et al. [134] used the modified Hamming function to represent the transmission path along the casing and also proposed two constants to model the transmission path inside the casing. Vanhollebeke et al. [227] used a combination of multi-body modeling and typical transfer path analysis [228] to investigate the impact of bearings on the total transfer path and the resulting vibration level.

6. Validation

Validation is a very important step in dynamics based fault diagnostics. Without proper and sufficient validations, theoretical results cannot be implemented in real applications and correspondingly generate marginal social and economic values. This section gives a review of methodology validation in three aspects, validation of gear mesh stiffness, dynamic models and fault diagnosis techniques.

6.1. Validation of gear mesh stiffness evaluation methods

Yau et al. [229] modeled a gear tooth in tapered shape, evaluated bending and shear deflection using a Rayleigh-ritz approach and validated their model by comparing with theoretical and experimental data. Cai [63] proposed a new mesh stiffness function for a pair of helical involute gear tooth pair and compared with theoretical calculation and experiment. Ma et al. [107] evaluated the mesh stiffness of a pair of spur gears with profile modification. Ma et al. [108] calculated the mesh stiffness of a spur gear pair considering tip fillet and friction. Ma et al. [126] evaluated the mesh stiffness of a pair of spur gears with spalling defect. Results in Refs. [107,108,126] are validated by comparing with finite element results, respectively. Wan et al. [109] proposed an accumulated integral potential energy method to evaluate the mesh stiffness of a pair of helical gears in healthy and cracked tooth conditions. Their method was validated by comparing with a finite element model and ISO standard 6336-1-2006 [230]. Rincon et al. [147] proposed a hybrid advanced model for gear mesh stiffness evaluation and compared with ISO standard 6336-1-2006. Refs. [101,103,119,124,116] investigated tooth crack effect on the mesh stiffness of a pair of gears using the potential energy method and validated their results by comparing with finite element results, respectively. Liang et al. [59] evaluated the mesh stiffness of a planetary gear set with tooth crack and validated their result by comparing with finite element results reported in [40]. Liang et al. [58] evaluated the mesh stiffness of a pair of spur gears with tooth pitting and validated their model using a linear finite element model. Gu et al. [69] presented approximate mesh stiffness formulae for solid spur and helical gears. Their results compared well with 2D finite element models and benchmark software codes. Parey et al. [161–163] built experiments to evaluate gear mesh stiffness and compared their results with an analytical model reported in [37] and finite element models. Wang and Zhang [130] validated their helical gear pair stiffness and stress calculation model with two reported models. Chabert et al. [231] validated their gear deflection model by ISO and AGMA standard.

6.2. Validation of gearbox dynamic models

This section aims to review validation methods of dynamic models that were mainly developed for the purpose of fault diagnosis. Cooley and Parker [31] gave a review of experiments on planetary gear dynamics and vibration. All the experimental studies reviewed in Ref. [31] will not be reviewed again.

A comprehensive mathematical model developed for analyzing the influence of shape deviations and mounting errors on gear dynamics was validated by comparing with several results from the literature [64]. A three-dimensional model of two-stage straight bevel gear was validated with a reported model and experimental studies for fault diagnosis [83]. A two-dimensional computational fluid dynamic model for wind energy system dynamic simulation was validated with experimental data [232]. Inalpolat and Kahraman [98] experimentally validated their planetary gearbox dynamic model

with manufacturing error considered. Guo and Parker [157] validated their planetary gearbox model with tooth wedging and bearing clearance nonlinearity using a finite element model. Wan et al. [118] correlated their cracked tooth dynamic model responses with experimental signals. Liang et al. [41] validated their planetary gear crack model by comparing simulated signals with experimental signals in time and frequency domains. Velez et al. [233] investigated the influence of profile modifications on narrow-faced spur and helical gear transmission errors and validated their model using a variety of reported models. Liu et al. [234] studied the effect of tooth friction and tooth bending on gear dynamics with validation by finite element results. Liu and Parker [235] analyzed the effect of tooth profile modification on multimesh gear vibration and validated their model against a finite element model and two existing models. James and Harris [236] proposed a 3D numerical model to predict load sharing between planets due to system deflections and manufacturing errors. Their results correlated with experimental test data very well. Vanhollebeke et al. [237] used experimental measurement results to validate the usability of a flexible multibody wind turbine gearbox model. He et al. [218] validated their vibro-acoustic model by comparing radiated sound pressure calculations with measured noise data over a range of operating torques. Gu and Velez [65] studied the influence of planet position errors on planetary gear dynamics and validated model using experimental evidence. Omar et al. [219] presented a 9DOF lumped parameter model of a one-stage gear system; experimental and simulated data were compared for different operating speeds, torque loads, and gear cracks. An analytical model [84] developed to simulate and analyze gear surface pitting and wear effect on vibration was evaluated using a gear fatigue test rig at NASA Lewis Research Center. Dynamic and phenomenological vibration models developed for failure prediction on planet gears of planetary gearboxes were validated with the experimental results extracted from two test benches [99]. Experimental data was employed to validate the dynamic characteristics of gears with local crack and spalling damage [120]. Yu et al. [199] validated their model by comparing simulated natural frequencies and dynamic transmission error with reported works. Hammami et al. [238] correlated a lumped-parameter model against experiments in modal analysis of a back-to-back planetary gearbox.

6.3. Validation of fault diagnosis techniques developed based on simulation

The validation of fault diagnosis techniques is to test the effectiveness of the techniques developed based on simulated signals on real systems. Cheng et al. [94,96] developed gear fault estimation approach for a planetary gearbox based on dynamic simulated vibration signals and grey relational analysis, and validated their method using experimental data. Parey et al. [178] tested empirical mode decomposition (EMD) pre-processed kurtosis and crest features on simulated and experimental signals to detect early tooth pitting fault. Bartelmus et al. [89] found that energy based statistical features like root mean square were load dependent in distributed fault detection and diagnosis based on gearbox dynamic analysis and experimental validation. Statistical features were used to detect gear tooth crack [113] and tooth spalling [70] based on simulated vibration signal analysis, and the statistical features' effectiveness was tested on experimental signals. Li et al. [140] investigated gearbox crack fault symptoms in frequency spectrum and sidebands using a multistage planetary gearbox dynamic model with experimental validation. Yesilyurt et al. [186] proposed to assess gear wear fault severity by measuring gear mesh stiffness reduction and validated their method using a gear test rig. Sikorska et al. [239] described a web-based management system for researchers to access suitable datasets for testing diagnostic and prognostic models. Yin et al. [173] tested their crack growth rate prediction model on a set of G6 gears. Endo et al. [172] validated their proposed differential diagnostic technique on a gear test rig. Ottewill et al. [195] validated their tooth profile fault monitoring technique by an experimental system simulating a wind turbine. Lei et al. [72] validated their carrier plate crack fault diagnosis method using a finite element model. Fault signatures found by Diehl and Tang's dynamic model [154] were validated by constant speed and varying speed experimental data. Liang et al. [106] proposed a windowing and mapping strategy to decompose planetary gearbox vibration signal into tooth levels which could enlarge fault symptoms and validated their modeled on a planetary gearbox test rig.

7. Summary and prospects

7.1. Summary

This paper mainly reviewed the following aspects: mesh stiffness evaluation, gear damage modeling and fault diagnosis techniques, gearbox transmission path modeling and method validations. Many researchers have attempted to obtain gearbox fault signatures from dynamic modeling and then develop effective fault detection and diagnosis techniques based on simulated signal analysis. Simulated gearbox signals do not have noise interference. Therefore, it is easy to observe fault signatures in simulated signals. But in the dynamic modeling, some simplifications are usually applied to reduce modeling difficulty and save computation cost. The simplifications may generate errors. To reduce simplification errors, sufficient comparisons and validations should be performed to make sure model accuracy.

The methods developed using dynamic modeling for gearboxes under constant operation conditions may not be suitable for gearboxes working under varying operation condition. However, the research findings under constant operation condition may provide theoretical foundation and also enlighten researchers for developing gearbox damage dynamic modeling methods under varying operation condition.

The main content of this paper is summarized as follows:

- Four methods for gear mesh stiffness evaluation are reviewed: square waveform method, potential energy method, finite element method, and experimental method. The square waveform method is simplest to use but cannot correlate with gear damage levels. The other three methods can correlate with gear damage levels. The potential energy method requires a deep knowledge in gear meshing theory. Users must be familiar with finite element modeling theory to utilize the finite element method. Several researchers designed experiments to evaluate gear mesh stiffness but it is time and cost consuming to design and build the experiments.
- Gearbox damage modeling and fault diagnosis techniques are reviewed for more than 7 failure modes. The research status for each failure mode is stated.
- The effect of transmission path and its modeling are reviewed for fixed-axis gearboxes and planetary gearboxes, respectively. For fixed-axis gearboxes, finite element method is basically used to analyze the transmission path effect. For planetary gearboxes, several window functions are mainly used to reflect distance changes between vibration sources and a fixed transducer due to the rotation of the carrier.
- Methods are reviewed for validation of gear mesh stiffness evaluation, gearbox dynamic modeling and fault diagnosis techniques, respectively.

Based on the scope of this review paper, some research perspectives are suggested in Section 7.2 for future consideration.

7.2. Research prospects

7.2.1. Evaluation of gear body effect on gear mesh stiffness in the potential energy method

As stated in Section 3.2, almost all researchers who considered the gear body effect used the Eq. (17) in Ref. [104] to reflect the gear body effect. However, this equation is only derived for a gear with a single tooth under load. If more than two pairs of teeth are in meshing, this equation may not be valid. In addition, this equation is derived for a perfect gear body. If gear damage like a crack goes inside the gear body, this equation is not valid.

7.2.2. Gear faults in multiple teeth of a gear: modeling and fault diagnosis techniques

For the failure modes of tooth cracking, tooth pitting/spalling, and tooth chipping, researchers basically only focus on a single tooth failure. In real applications, failure may occur in multiple teeth simultaneously. Therefore, it is necessary to model gear faults in multiple teeth of a gear and develop fault diagnosis techniques.

7.2.3. Gear faults in multiple gears: modeling and fault diagnosis techniques

Inside a gearbox, there are possible many gears. Current research focuses on a failure on a single gear by assuming all other gears are perfect. In real applications, different degree of failures may happen in multiple gears. Modeling and investigating multiple gear faults can better meet real application requirements.

7.2.4. Gear mesh damping effect analysis: modeling and fault diagnosis techniques

Current lumped parameter models mostly assumed gear mesh damping ratio was a constant and independent of gear damage. In real applications, gear mesh damping may not be constant for different tooth contact positions. In addition, gear damage effect on mesh damping may deserve some investigations.

7.2.5. Dynamic modeling of gearbox faults under varying operation condition

As we know, gearboxes are common to work under varying operation condition. Some studies [240–249] modeled gearbox transmission systems under varying operation condition but the systems do not have damages involved. Very limited work has been done on gearbox damage dynamic modeling under varying operation condition. Based on my literature review, only four papers [89,209–211] modeled gearbox damage under varying operation condition. More studies are required to model gear faults under varying operation condition.

7.2.6. Effects of noises from internal and/or external sources

Simulated vibration signals do not have noise interference. The noise could be from the environment or from some other components in a gear transmission system. The noise effect should be modeled and investigated in developing fault diagnosis techniques.

7.2.7. Comprehensive modeling of gearbox degradation process

Bartelms [4,86] suggested that the researchers needed to be familiar with the gearbox's driving schemes including system of elements (engine, gear, bearing, shaft etc.), influencing factors (design, production, operation and condition change), interaction of gearbox elements and also with the working environment, and component damage mechanism in order to develop a robust condition monitoring and diagnostic method for gearboxes, especially for complex gearboxes with multi-stages. One degradation scenario given in Ref. [4] was the influence of hard dust particles on roller element bearings (REB) causing the increase of inner backlash of REBs. The increased inner backlash caused shafts misalignment and

misalignment in gear meshing. Consequently, this state of gear cooperation caused other faults such as spalling, pitting, crack and breakage. However, these factors mentioned in Ref. [4] have not been covered by the reported dynamics modeling work. It is a good practice to consider these factors in the future gearbox degradation modeling.

7.2.8. Experimental validation of models for transmission path effects

Some models have been developed to investigate the transmission path effects. However, rare experimental validation is conducted.

7.2.9. Repeatability tests and test on multiple machines to develop robust fault diagnosis techniques

Some dynamics based fault diagnosis techniques are tested to be effective on one set of data or one experimental test rig. Repeatability tests and tests on multiple machines are required for these techniques to be used in real applications.

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